

# 李伟凯 学术简历

教授 / 博士生导师 / 泰山学者青年专家

leeweikai@sdjzu.edu.cn / Google Scholar: scholar.google.com/citations?user=XEfV8mkAAAAJ

山东建筑大学 计算机与人工智能学院 / 重庆交通大学 数学与统计学院 / 全景医学影像中心



## 教育经历

- 2018 – 2022 / 博士 / 计算机科学与技术 / 南京航空航天大学
- 2015 – 2018 / 硕士 / 计算机技术与应用 / 重庆交通大学
- 2011 – 2015 / 学士 / 信息与计算科学 / 重庆交通大学

## 科研项目

- 域对齐外的无监督域适应研究及其拓展 / 国家自然科学基金青年项目 / 主持
- Advanced School in Applied Machine Learning / 国家自然科学基金国际合作交流项目 / 主持
- 抑郁症异质性脑网络的多模态神经生理机制解析与数字孪生闭环经颅磁刺激机器人关键技术研究 / 山东省重大基础研究项目 / 主持
- 域对齐外的无监督域适应探索 / 工信部重点实验室开放项目 / 主持
- 基于 PET/MR 多模态脑网络的帕金森症早期诊断及识别 / 分子影像湖北省重点实验室开放项目 / 主持
- 基于多模态影像的自闭症多尺度模式分析与研究 / 重庆市教委科研项目青年基金 / 主持
- 基于多中心迁移学习的脑网络疾病诊断方法研究 / 重庆市自然科学基金面上项目 / 主持
- 调度协议和网络攻击下随机复杂网络的保概率同步性及安全控制研究 / 重庆市科技局自然科学基金创新发展联合基金 / 主研
- 自监督学习本质的有效性探究、论证及拓展增强 / 国家自然科学基金面上项目 / 主研

## 论文成果

- 2026 / Universum Inspired Graph Contrast Learning for Major Depressive Disorder Identification: In Functional Brain Network View. H Qin, G Hong, D Ma, Z Wang, L Zhao, W Li, X Xu. CNS Neuroscience & Therapeutics 32 (9), e71158, 2026
- 2026 / Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI. Z Zheng, X Chen, H Jiang, R Tian, Q Cai, J Liu, X Chen, W Li, Y Wang. arXiv preprint arXiv:2607.29531, 2026
- 2026 / Domain generalization meets mutual bi-attention: A novel approach to drug-target prediction. J Yang, W Li, K Qian, Z Cheng. Neurocomputing, 134626, 2026
- 2026 / Enhancing Fatigue Detection Through Heterogeneous Multisource Data Integration and Cross-Domain Modality Imputation. L Cui, Y Wu, Y Tang, W Li. IEEE Transactions on Computational Social Systems, 2026
- 2026 / Hierarchical neighbor integration graph attention network for autism spectrum disorder diagnosis. D Ma, L Peng, L Zhang, W Li, X Gao. Frontiers in Psychiatry 17, 1846996, 2026
- 2026 / TumorAL: Evidence-aware active learning for 3D tumor segmentation. H Wang, J Leng, S Li, Y Zhao, W Li, W Li, X Gao. Neurocomputing, 134239, 2026
- 2026 / Identification of Major Depressive Disorder Using Multiple Functional Connection Patterns. Y Ruan, L Guan, L Peng, L Zhao, Y Huang, W Li, X Gao. CNS Neuroscience & Therapeutics 32 (6), e70951, 2026
- 2026 / Electrocardiographic Alterations Combined with Hematological, Biochemical, and Metabolic Profiles Predict Prognosis in Kawasaki Disease. Q Wang, W Li, J Wan, L Wei, Y Xia, Y Hua, K Zhou, D Qie, W Li, Y Li. Journal of Cardiovascular Development and Disease 13 (6), 228, 2026

9. 2026 / Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq. X Chen, Q Zou, Q Cai, X Chen, W Li, Y Wang. *PLOS Computational Biology* 22 (5), e1014223, 2026
10. 2026 / Bridging the modality reliability gap in drug-target interaction prediction via a confidence-aware multimodal fusion framework. J Yang, J Zhang, K Qian, Q Yang, W Li, Z Cheng. *Proceedings of the AAAI Conference on Artificial Intelligence* 40 (32), 27529 ..., 2026
11. 2026 / Test-time few-shot object detection via dynamic prototype fusion. Y Wu, Y Li, H Wei, W Li, Y Tang. *IEEE Transactions on Cybernetics*, 2026
12. 2026 / Machine learning-based assessments of perioperative features in determining long-term conduct block post-transcatheter closure of ventricular septal defect. Q Wang, Y Xia, L Wei, P Yang, K Zhou, Y Hua, W Li, Y Li. *Cardiology in the Young*, 1-11, 2026
13. 2025 / Attention Transfer Based Hybrid Knowledge Distillation for Multimodal Brain Tumor Segmentation. H Ma, HS Gan, H Mi, P Xu, Y Liu, W Li. 2025 *IEEE International Conference on Big Data (BigData)*, 7503-7510, 2025
14. 2025 / More insights into disruption and decoupling of individual metabolic connectomes in Parkinson's disease. D Li, J Yao, W Li, Z Guo, X Lv, L Chen, J Shi, YC Chen, J Ye. *Progress in Neuro-Psychopharmacology and Biological Psychiatry*, 111578, 2025
15. 2025 / Incorporating Negative Node Information in Graph Transformer for MDD Diagnosis. H Song, H Qin, W Li. 2025 *International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
16. 2025 / MVSSL-GT: A Graph Transformer Method with Multi-View Contrastive Self-Supervised Learning. Y Lu, N Tian, W Li. 2025 *International Conference on Cyber-Physical Social Intelligence (CPSI)*, 1-6, 2025
17. 2025 / Low-light image enhancement using dual cross attention. Y Ruan, H Ma, D Ma, W Li, X Wang. *Engineering Applications of Artificial Intelligence* 159, 111501, 2025
18. 2025 / Auto-weighted projective one-step multi-view clustering. X Mou, W Li, B Li, J Hu. *Neurocomputing*, 131960, 2025
19. 2025 / Revisiting Multi-Modal Alignment: In Distribution View. W Li, N Tian, Y Li, Y Tang. 2025 *IEEE International Conference on Systems, Man, and Cybernetics (SMC)* ..., 2025
20. 2025 / Quantitative susceptibility mapping shows alterations of brain iron content in children with autism spectrum disorder: a whole-brain analysis. X Xu, Y Li, H Lan, N Ding, W Li, G Zheng, X Song. *BMC psychiatry* 25 (1), 826, 2025
21. 2025 / Cross-modal medical image generation from MRI to PET using robust generative adversarial network. Y Yang, B Li, W Cao, X Chen, W Li. *Expert Systems with Applications*, 129287, 2025
22. 2025 / Leveraging multi-source and heterogeneous signals for fatigue detection. L Cui, Y Wu, T Ying, W Li. *arXiv e-prints*, arXiv: 2507.16859, 2025
23. 2025 / PSMA PET/CT based multimodal deep learning model for accurate prediction of pelvic lymph-node metastases in prostate cancer patients identified as candidates for extended ... Q Ma, B Chen, R Seifert, R Zhou, L Xiao, J Yang, A Rominger, K Shi, W Li, .... *European Journal of Nuclear Medicine and Molecular Imaging* 52 (6), 2063-2075, 2025
24. 2025 / Timestamp calibration for time-series single cell rna-seq expression data. X Chen, S Lin, X Chen, W Li, Y Li. *Journal of Molecular Biology* 437 (9), 169021, 2025
25. 2025 / Iterative neural networks for improving memory capacity. X Chen, D Lin, Z Li, W Li. *Neural Networks* 182, 106936, 2025
26. 2024 / The hidden Markov model reveals the changes in brain dynamics among patients with end-stage renal disease under different dialysis methods. D Sun, W Li, H Yu, X Chen, C Zhang. *Journal of Cyber-Physical-Social Intelligence* 3 (1), 10-10, 2024
27. 2024 / An Improved Algorithm for Spiking Neural Networks with Multi-Scale Attention Coding. S Chen, X Chen, W Li. 2024 *International Conference on Cyber-Physical Social Intelligence (ICCSI)*, 1-6, 2024
28. 2024 / Improved domain-adversarial neural networks with mixup and pseudo-labeling. X Xiao, X Chen, W Li. 2024 *International Conference on Cyber-Physical Social Intelligence (ICCSI)*, 1-6, 2024
29. 2024 / Dynamic Graph Contrastive Learning for Neuro Disease Identification using Resting-State fMRI. Y Jiang, W Li, X Chen. 2024 *International Conference on Cyber-Physical Social Intelligence (ICCSI)*, 1-6, 2024
30. 2024 / Synthesizing A  $\beta$  -PET based on multi-modal neuroimaging fusion for pathological diagnosis of Alzheimer ' s disease. Y Yang, B Li, W Li, H Chen, W Cao. 2024 *International Conference on Cyber-Physical Social Intelligence (ICCSI)*, 1-6, 2024
31. 2024 / Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification. H Qin, X Xu, W Li. 2024 *International Conference on Cyber-Physical Social Intelligence (ICCSI)*, 1-6, 2024

32. 2024 / Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization. D Sun, X Chen, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
33. 2024 / Multi-view comparative self-supervised graph learning for neuro-disease diagnosis using functional brain networks. N Tian, X Xu, W Li. 2024 International Conference on Cyber-Physical Social Intelligence (ICCSI), 1-6, 2024
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44. 2024 / Effects of individual metabolic brain network changes co-affected by T2DM and aging on the probabilities of T2DM: protective and risk factors. YL Li, JJ Wu, WK Li, X Gao, D Wei, X Xue, XY Hua, MX Zheng, JG Xu. *Cerebral Cortex* 34 (1), bhad439, 2024
45. 2024 / Topological properties analysis and identification of mild cognitive impairment based on individual morphological brain network connectome. X Xu, P Chen, W Li, Y Xiang, Z Xie, Q Yu, Y Tang, P Wang. *Cerebral Cortex* 34 (1), bhad450, 2024
46. 2023 / Wavelet transform-based frequency self-adaptive model for functional brain network. Y Ding, X Xu, L Peng, L Zhang, W Li, W Cao, X Gao. *Cerebral Cortex* 33 (22), 11181-11194, 2023
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53. 2023 / Clinical application of machine learning methods in psychiatric disorders. X Liu, Z Xu, W Li, Z Zhou, X Jia. *Frontiers in psychiatry* 14, 1209615, 2023
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55. 2022 / Unsupervised domain adaptation with progressive adaptation of subspaces. W Li, S Chen. *Pattern Recognition* 132, 108918, 2022

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57. 2022 / Machine learning-derived multimodal neuroimaging of presurgical target area to predict individual's seizure outcomes after epilepsy surgery. Y Tang, W Li, L Tao, J Li, T Long, Y Li, D Chen, S Hu. *Frontiers in Cell and Developmental Biology* 9, 669795, 2022
58. 2022 / Human-Guided Functional Connectivity Network Estimation for Chronic Tinnitus Identification: A Modularity View. WK Li, YC Chen, XW Xu, X Wang, X Gao. *IEEE journal of biomedical and health informatics*, 2022
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62. 2021 / Morphological, structural, and functional networks highlight the role of the cortical-subcortical circuit in individuals with subjective cognitive decline. X Xu, T Wang, W Li, H Li, B Xu, M Zhang, L Yue, P Wang, S Xiao. *Frontiers in Aging Neuroscience* 13, 688113, 2021
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64. 2020 / Leave zero out: Towards a no-cross-validation approach for model selection. W Li, C Geng, S Chen. arXiv preprint arXiv:2012.13309, 2020
65. 2020 / Effective and accurate diagnosis of subjective cognitive decline based on functional connection and graph theory view. X Xu, W Li, M Tao, Z Xie, X Gao, L Yue, P Wang. *Frontiers in neuroscience* 14, 577887, 2020
66. 2020 / Functional connectivity network estimation with an inter-similarity prior for mild cognitive impairment classification. W Li, X Xu, W Jiang, P Wang, X Gao. *Aging (Albany NY)* 12 (17), 17328, 2020
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75. 2018 / 一种自适应分数阶偏微分图像增强模型.. 李伟凯, 王政霞, 蒋伟. *Computer Engineering & Science/Jisuanji Gongcheng yu Kexue* 40 (4), 699, 2018
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79. — / An Improved Efficient Algorithm for Spiking Neural Network Through Dataset Distillation. S Chen, X Chen, W Cao, W Li. Available at SSRN 4836784
80. — / Graph Adversarial Discriminative Domain Adaptation with Category-Level Alignment. X Xiao, X Chen, W Li, Z Li, W Cao, S Zhang. Available at SSRN 4898772

## 学术服务

1. CPSI 2027 程序主席
2. CPSI 2026 联合程序主席
3. IEEE SMC 2026 领域主席
4. IFAC CPHS 2020 领域主席
5. ICCSI 2022 – 2025 领域主席
6. 具身智能系统及应用大会平行论坛召集人
7. 《数据采集与处理》青年编委
8. 《Meta-Radiology》青年编委
9. 《Frontiers in Neuroscience》客座主编
10. 《Frontiers in Aging Neuroscience》客座主编
11. 《Frontiers in Cell and Developmental Biology》客座主编
12. 《Frontiers in Pharmacology》客座主编
13. 《数据采集与处理》优秀审稿人
14. 《Aging & Disease》青年编委
15. 《Brain-X》青年编委
16. 《Cog》青年编委
17. 《Artificial Intelligence Science and Engineering》青年编委
18. 中国人工智能学会机器学习专委会通讯委员
19. 中国指挥与控制学会具身智能专委会副秘书长
20. 中国自动化学会人机教育专委会委员
21. 中国仿真学会虚拟技术及应用专委会委员
22. 医学图像青年研讨会委员
23. 重庆市数学学会理事

## 个人荣誉

1. 2026 / CPSI 最佳论文奖
2. 2025 / 泰山学者青年专家
3. 2025 / CPSI 最佳论文奖
4. 2025 / 中国机器人及人工智能大赛优秀指导教师
5. 2023 / ICCSI 最佳论文入围奖
6. 2022 / 博士研究生国家奖学金

## 指导学生获奖

1. 2026 / 崔靖轩等获 CPSI 2026 最佳论文奖
2. 2025 / 中国机器人及人工智能大赛全国总决赛一等奖

3. 2025 / 中国机器人及人工智能大赛重庆选拔赛一等奖
4. 2025 / 全国数字艺术设计大赛全国三等奖
5. 2025 / 美国大学生数学建模竞赛 H 奖、M 奖
6. 2025 / 中国国际大学生创新大赛校赛一等奖：帕影先知
7. 2025 / 中国国际大学生创新大赛校赛一等奖：影络智诊
8. 2025 / iCAN 校赛二等奖
9. 2025 / 中国研究生电子设计竞赛西南赛区三等奖
10. 2024 / 美国大学生数学建模竞赛 H 奖、M 奖
11. 2024 / 中国国际大学生创新大赛校赛一等奖、三等奖
12. 2024 / 中国机器人及人工智能大赛全国总决赛二等奖
13. 2024 / 机器人竞赛重庆市二等奖
14. 2024 / 全国高校商业精英挑战赛全国总决赛一等奖
15. 2024 / 全国高校智能交通创新与创业大赛三等奖
16. 2024 / 中国研究生电子设计竞赛三等奖
17. 2024 / 校级创业大赛二等奖
18. 2024 / 睿抗机器人开发者大赛一等奖
19. 2023 / 全国大学生数学建模竞赛重庆赛区一等奖及二等奖
20. 2023 / 中国 ‘ 互联网+ ’ 大学生创新创业大赛校赛三等奖
21. 2023 / 中国软件杯西部赛区三等奖